CS 3600 Project Phase 2

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StockUp

Contents

1 Project

StockUp is a social and financial app which is a combination of fantasy football and real-world investment. On StockUp, users will be able to join a team, invest in stocks, and earn money for their team.

2 System Design

2.1 Database

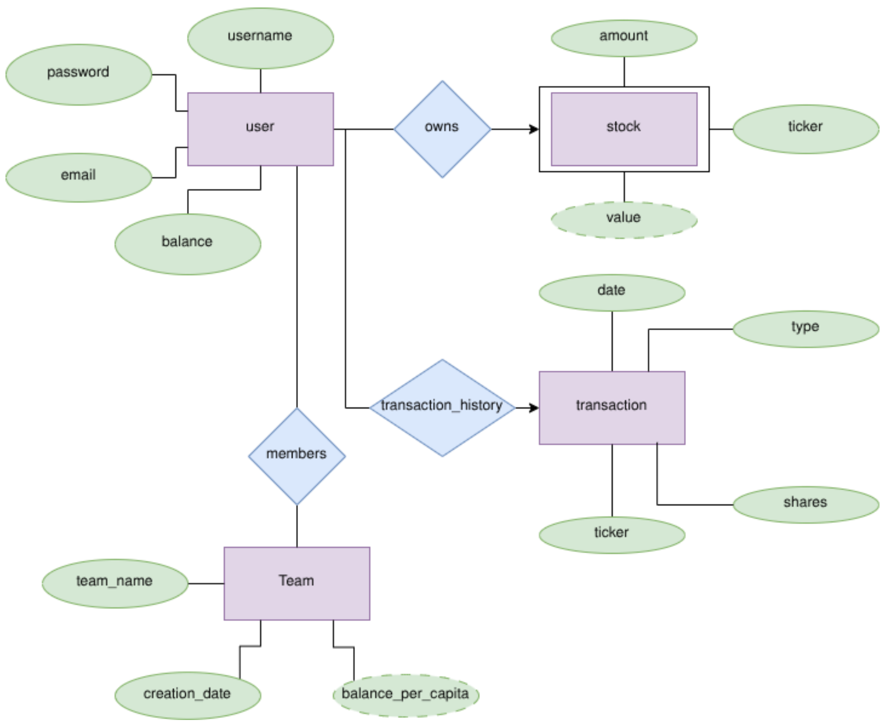

Figure 1: ER Diagram

Django and SQLite3 are used to create and query the database. The tables are generated by Django with the models.py file.

Any changes made to the database model are applied by executing the migrate command with manage.py.

2.2 Backend

The backend routes are routed by Django. All routes are implemented in the views.py file. The routing URLs are assigned in the urls.py file from implementations in the views.py file.

2.3 Frontent

The frontend is designed with HTML and Bootstrap templates generated with Django. Templates are stored in the templates directory with static resources from the static directory. Essentially, HTML files are stored in the templates directory and CSS/JS files are stored in the static directory for sharing.

2.4 Stock Predictor

The stock information is gained from the Yahoo Finance package, yfinance. The stock predictor model is implemented with XGBoost and stored as a file that is restored with the joblib package. The joblib package allows for the model to be trained once to reduce prediction times.

A prediction may be obtained through a POST request sent to the invest route. The POST request body requires a ticker symbol and a sector.

3 Functional Components

3.1 Database

The database is fully implemented.

3.2 Backend

The core features of the backend are implemented with routes, database, and pages.

3.3 Frontent

User interface is complete and the content is implemented. The current interface is clean and modern. However, additional refining is desired for improved ergonomics.

3.4 User Login

The user login and sign up forms allow a user to sign into their account or create a new account with their email address and desired password.

3.5 Stock Predictor

The stock predictor is partially implemented. The user may search a particular stock to view the average price history of the stock over a course of the past one-hundred days. Afterwards, the user may select the stock industry to tune the predictor to more accurately predict the selected stock. Once the information is submitted, a graph appears that shows the stock price history along with the predicted price in the next thirty days to help the user to gauge whether the stock is worth purchasing. After a decision is made by the user and the invest option is selected, the user is charged and the stock becomes owned by the user's StockUp account.

3.6 Leaderboards

The leaderboards show the top ranking teams based on the value per capita. In the team view, users may fill a search bar text box and submit the search result to view teams that match the specified search. Each element in the list of teams shows the team's name, number of members, and the balance per capita. The user may select a team to view its details.

3.7 Team View

When viewing the details of a team, the user is presented with the team's number of members, balance per capita, and total balance. Additionally, a graph shows each member of the team displayed as a node with name that is connected to the team node. The user may request to join the team by selecting the join team option.

3.8 Administration Utilities

There are administration tools that allow the server administrators to view and modify database values and tables. The utilities are protected with an admin login that requires user privileges to access. These utilities allow for easy debugging and error tracking to resolve any potential issues.

4 Validation

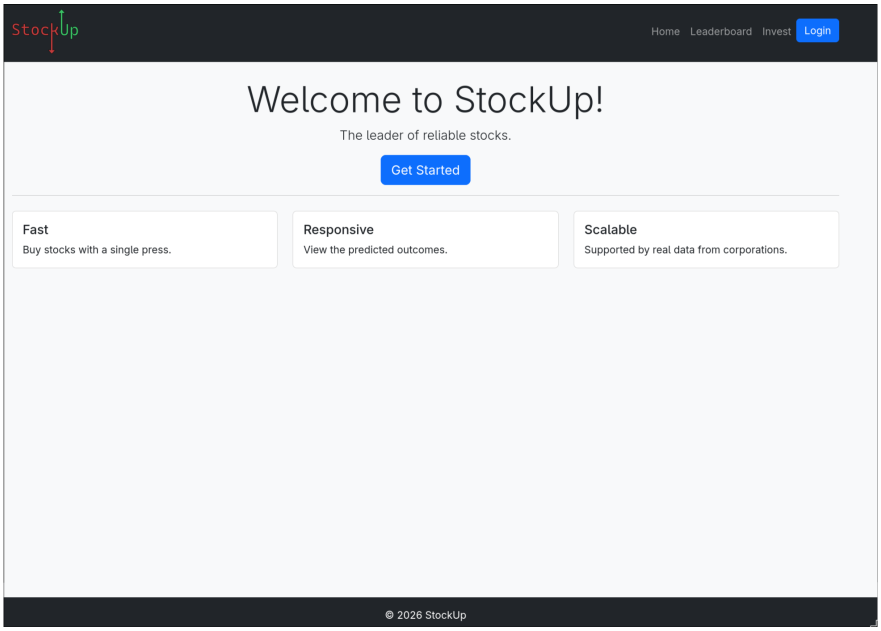

Figure 2: Validation Screenshot

Figure 3: Stock Investing

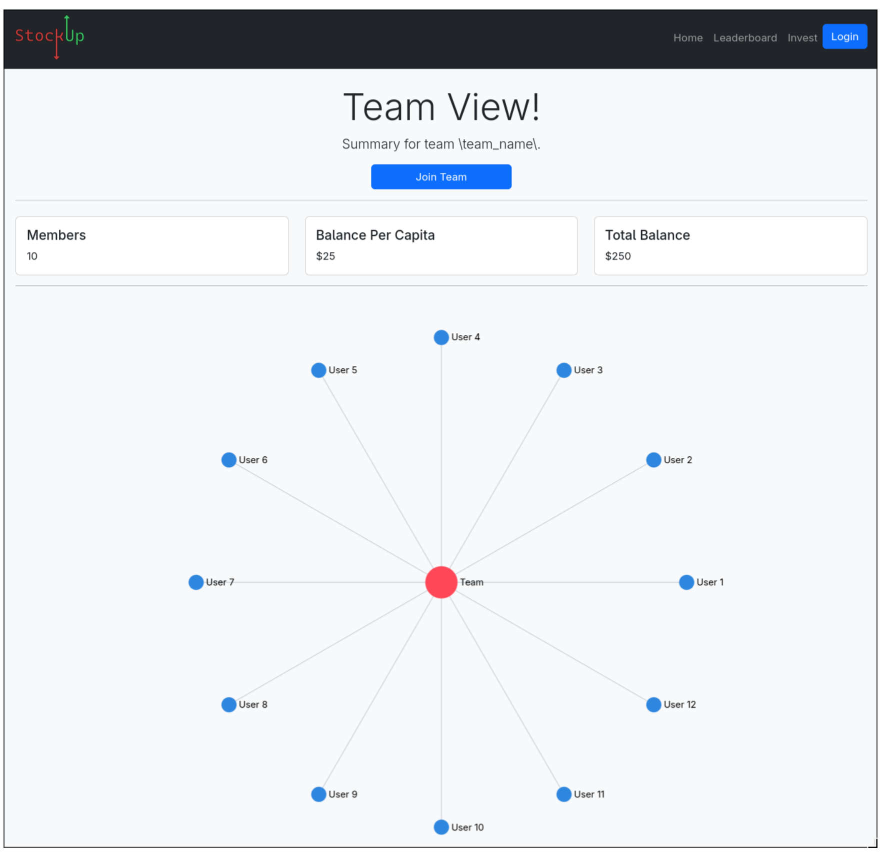

Figure 4: Team View

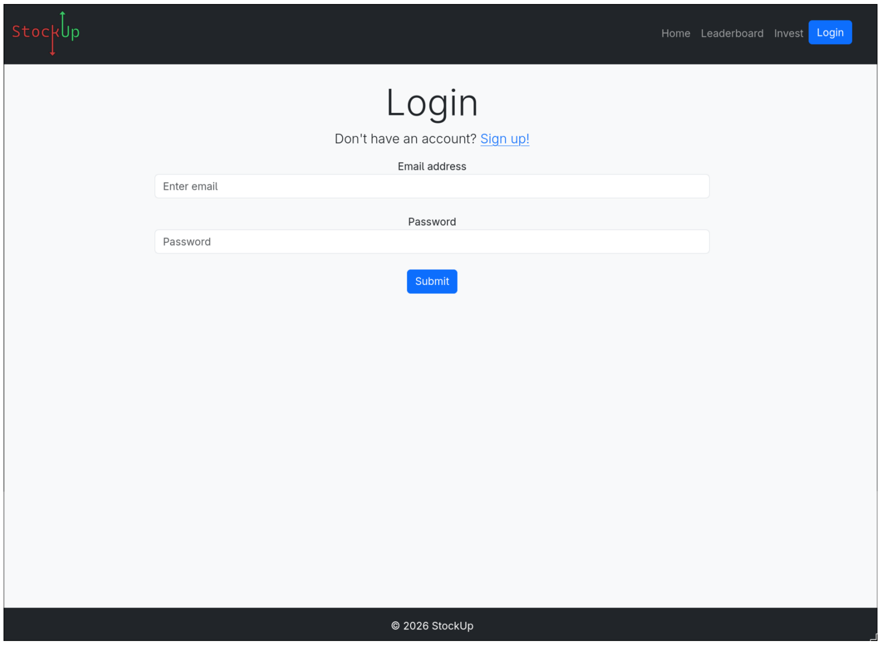

Figure 5: Login

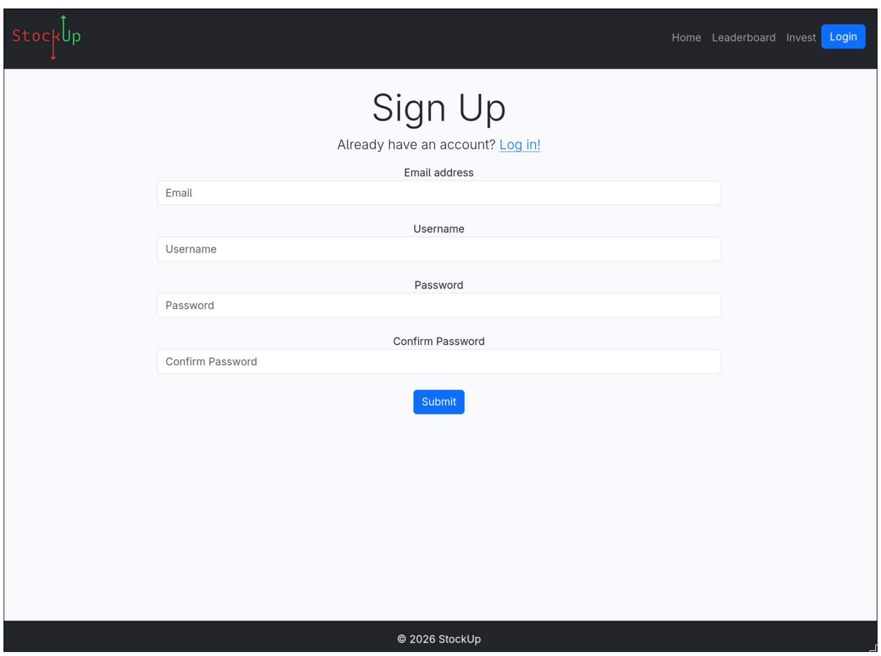

Figure 6: Sign Up

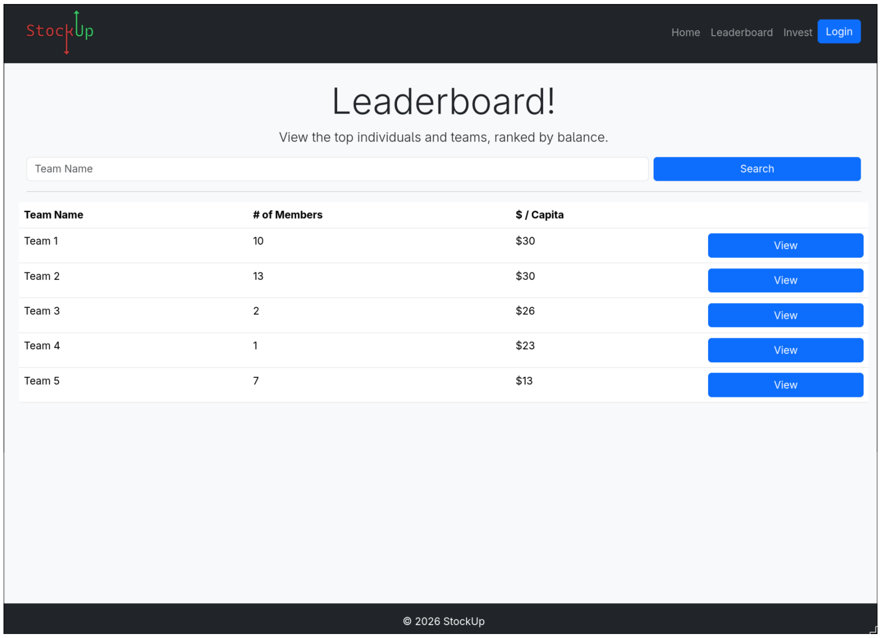

Figure 7: Leaderboards

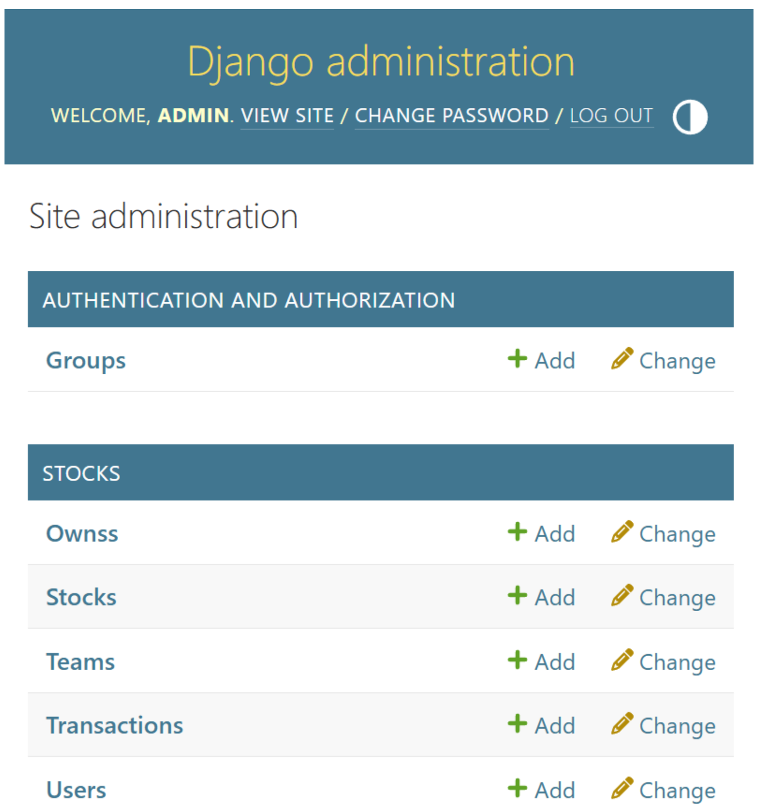

Figure 8: Database Administration

5 Timeline of Deliverables

- **April 15, 2026:** All parts of the project must work in isolation of each other. The website should be fully functional, the backend and API should be fully functional, and the database should be fully functional. At this point, the project should be done with Phase III.
- **May 1, 2026:** The project should be fully functional at this point. The required queries should be in place, and all aspects of the project should properly interact with minimal bugs. Phase IV should be complete at this point.
- **May 15, 2026:** The project must be complete by this point.